Structure as Constraint:

A Formal Framework for Non-Derivability Across Domains

Abstract

We introduce a constraint-based formal framework in which *structure* is defined as a non-ontological operator limiting the space of derivable trajectories across explanatory domains. Physics and metaphysics are modeled as orthogonal descriptive spaces whose relation is fixed by structural constraints rather than causal reduction. We formalize the distinction between internally derivable systems (N_0) and non-derivable rupture regimes (N_1), and show that systemic collapse corresponds to the invalidation of internal dynamics rather than stochastic deviation. The framework explains why historical and evolutionary processes are locally derivable but globally fragile, and why intelligibility is fundamentally retrodictive.

1. Introduction

Physics and metaphysics are often treated as rival explanatory frameworks. One claims authority over measurable phenomena; the other over necessity, meaning, or ultimate grounding. Despite centuries of debate, neither subsumes the other.

We argue that this is not a failure of either domain, but a consequence of a missing formal layer: *structure*.

Structure is not a third ontology. It is not a substance, a cause, or a hidden realm. Instead, structure functions as a constraint operator that limits which configurations, transitions, and continuities are admissible across domains. This paper develops a formal framework for structure as a boundary condition on derivability.

2. The Problem of Closure and Derivability

Most scientific and philosophical explanations implicitly assume *closure*: the idea that, given sufficient information, future states remain derivable from past states.

This assumption underwrites prediction, optimization, and narratives of progress. However, collapse events — extinctions, systemic failures, regime shifts — demonstrate that derivability can fail even in deterministic systems.

We therefore distinguish between:

- **Local derivability** within a valid regime
- **Global derivability** across regime invalidation

We argue that the latter is structurally impossible.

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3. Explanatory Domains

3.1 Physics and Metaphysics

Definition 1 (Explanatory Domains).
Let P denote a space of physical descriptions (laws, states, measurable quantities), and M a space of metaphysical descriptions (categories, necessity, meaning).

We assume:

$$P \ncong M, \qquad P \nsubseteq M, \qquad M \nsubseteq P$$

Neither domain reduces to the other.

3.2 Orthogonality

Define the explanatory space:

$$E = P \oplus M$$

equipped with an inner product $\langle \cdot, \cdot \rangle$ such that:

$$\langle P, M \rangle = 0$$

This formalizes the claim that physical and metaphysical explanations are orthogonal: they do not negate, subsume, or resolve one another.

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4. Structure as a Constraint Operator

4.1 Definition of Structure

Definition 2 (Structure).
Let $\Omega = P \times M$ be the space of explanatory configurations.
Structure is a function:

$$S : \Omega \to \{0, 1\}$$

where:

- $S(\omega) = 1$ denotes an admissible configuration
- $S(\omega) = 0$ denotes an inadmissible configuration

Structure does not generate states or dynamics. It filters the space of possibilities.

4.2 Structure Is Not a Dynamic

Theorem 1 (Categorical Non-Equivalence of Structure and Dynamics)

Let:

- X be a state space
- $F : X \to X$ a state-transition function (dynamic)
- $\Omega = P \times M$ a configuration space
- $S : \Omega \to 0, 1$ a structural constraint operator

Then **there exists no function** F such that S is equivalent to F under any isomorphism preserving domain and codomain structure.

Formally:

$$S \not\cong F$$

for all admissible $F : X \to X$.

Proof

Assume, for contradiction, that there exists a state space X and a transition function $F : X \to X$ such that S is equivalent to F.

By definition of a dynamic, F satisfies:

$$\forall x \in X, \quad F(x) \in X$$

That is, F is **total and closed** over X.

Now consider the structure operator $S : \Omega \to 0, 1$.
S does **not** map configurations to configurations, but instead assigns **admissibility**:

$$S(\omega) = \begin{cases} 1 & \text{if } \omega \text{ is admissible} \\ 0 & \text{otherwise} \end{cases}$$

Hence:

- the codomain of S is **not** Ω
- S does **not** preserve membership in any state space
- S does **not** define a successor relation

Moreover, there exist configurations $\omega \in \Omega$ such that:

$$S(\omega) = 0$$

i.e., configurations that are **structurally forbidden** and therefore have **no successor state at all**.

This violates the defining property of any dynamic F, which must be defined on all elements of its domain.

Therefore, no function $F : X \to X$ can be equivalent to S.

This contradicts the initial assumption.

Hence, structure and dynamics are categorically distinct.

4.3 Projection Without Reduction

There exist projection maps:

$$\pi_P : \Omega \to P, \qquad \pi_M : \Omega \to M$$

However:

$$\pi_P^{-1} \nexists, \qquad \pi_M^{-1} \nexists$$

Thus, structure manifests in both domains without being reducible to either.

4.4 Corollaries of Structural Non-Equivalence

The categorical non-equivalence between structure and dynamics established in Theorem 1 yields several immediate and non-trivial consequences. These corollaries clarify the epistemic, temporal, and explanatory limits imposed by structural constraints.

Corollary 1 (Pre-Dynamical Status of Structure)

Structure operates on the space of possible configurations *prior* to and *independently of* any state-transition dynamics.

Formally, for any dynamic $F : X \to X$ and any configuration $\omega \in \Omega$:

$$S(\omega) \text{ is defined independently of } F$$

Dynamics presuppose admissibility; structure determines it.

Corollary 2 (Impossibility of Structural Prediction)

No structural constraint can be inferred or predicted from internal dynamics alone.

That is, for any dynamic $F : X \to X$:

$$S \notin \sigma(F)$$

where $\sigma(F)$ denotes the set of properties inferable from F.

Structural boundaries cannot be extrapolated from trajectories, even under perfect information and deterministic evolution.

Corollary 3 (Non-Dynamical Nature of Collapse)

An N_1 event does not correspond to a failure or breakdown of dynamics, but to the invalidation of the domain on which the dynamics is defined.

Formally:

$$N_1 \neq \neg F \quad \text{but} \quad N_1 = \neg \mathrm{dom}(F)$$

Collapse is therefore not a dynamical anomaly, but a structural boundary.

Corollary 4 (Global Non-Closure)

Even if an N_0 system is locally closed and deterministic, the global configuration space Ω is not closed under derivation.

Formally:

$$\forall F : X \to X, \quad \exists \omega \in \Omega \text{ such that } S(\omega) = 0$$

Local closure does not imply global closure.

Corollary 5 (Retrodictive Asymmetry)

After an N_1 event, internal forward derivability is impossible.
Explanatory coherence, if any, can only be reconstructed retrospectively from admissible pre-rupture states, and not extended forward by internal derivation.

Remark (Retrodiction vs. Inverse Dynamics)

Retrodiction in this framework does **not** imply the existence of an inverse or backward dynamic F^{-1}.
The claim of retrodictive coherence refers solely to the *explanatory reconstruction* of admissible pre-rupture states within an N_0 regime. No dynamic extension—forward or backward—is defined across an N_1 event.

Thus, retrodiction is an epistemic operation on already admissible histories, not a temporal inversion of the system's dynamics.

Summary

These corollaries collectively establish that structural constraints:

- are not dynamically derivable,
- are not temporally symmetric,
- and impose hard limits on prediction, optimization, and continuity.

They transform the non-equivalence of structure and dynamics from a conceptual distinction into a set of formal consequences with direct implications for scientific and philosophical explanation.

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5. The N_0 / N_1 Distinction

5.1 N_0: Internally Derivable Regimes

Definition 3 (N_0 system).
An N_0 system is a triple:

$$N_0 = (X, T, F)$$

where:

- X is a state space
- T is an internal time parameter
- $F : X \to X$ is a derivable dynamic

For all $x_t \in X$:

$$x_{t+1} = F(x_t)$$

Within N_0, prediction and optimization are meaningful.

5.2 N_1: Rupture and Invalidity

Definition 4 (N_1 event).
An N_1 event occurs if there exists t^* such that:

$$F(x_{t^*}) \notin X$$

or equivalently, F ceases to be well-defined.

N_1 events do not modify trajectories; they invalidate them.

5.3 Structure as Boundary

Proposition 2.
Structure constitutes the boundary between N_0 and N_1:

$$S = \partial N_0$$

Structure determines the conditions under which derivability holds and the thresholds beyond which it collapses.

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6. Non-Derivability Across Rupture

We now formalize the claim that derivability cannot be extended across structural rupture. This result follows necessarily from the distinction between internally derivable regimes (N_0) and structural invalidation events (N_1).

Theorem 2 (Non-Derivability Across Structural Rupture)

Let $N_0 = (X, T, F)$ be an internally derivable system, where $F : X \to X$ is a well-defined state-transition function.

Let $t^* \in T$ be such that an N_1 event occurs at t^*, i.e.:

$$F(x_{t^*}) \notin X$$

Then there exists **no extension** $F' : X \to X$ such that:

1. $F'(x) = F(x)$ for all $x \in X$ with $t < t^*$, and

2. F' remains well-defined for $t \geq t^*$.

Therefore, no trajectory defined within N_0 can be extended across an N_1 event by internal derivation.

Proof

Assume, for contradiction, that such an extension $F' : X \to X$ exists.

By condition (1), F' coincides with F on all admissible states prior to t^*.
By condition (2), F' is total and closed over X for all $t \geq t^*$.

Since F' is a state-transition function, it must satisfy:

$$\forall x \in X, \quad F'(x) \in X$$

In particular, this must hold for $x_{t^*} \in X$, implying:

$$F'(x_{t^*}) \in X$$

But by definition of an N_1 event:

$$F(x_{t^*}) \notin X$$

and since F' agrees with F up to t^*, any such F' would require redefining the image of x_{t^*} to remain within X.

This contradicts the definition of N_1 as a structural invalidation of the domain on which F is defined.
Such a redefinition would not constitute an extension of the original dynamics, but a change of regime.

Therefore, no such extension F' exists.

Hence, derivability across an N_1 event is impossible.

Corollary 6 (Irreversibility of Structural Rupture)

Once an N_1 event occurs, any subsequent dynamics necessarily belongs to a different regime $N_0' = (X', T', F')$ with:

$$X' \neq X$$

Structural rupture is therefore irreversible with respect to the original state space.

6.1 Meta-Theorem: Structural Limits of Derivability

We now state a meta-theorem connecting the categorical non-equivalence of structure and dynamics (Theorem 1) with the impossibility of derivation across rupture (Theorem 2). The result is stated as a **unidirectional implication**, which is sufficient for the framework and avoids unnecessary strengthening.

Meta-Theorem (Structural Origin of Non-Derivability — Refined)

Let $\Omega = P \times M$ be a configuration space equipped with a structural constraint operator $S : \Omega \to 0, 1$.

Let $N_0 = (X, T, F)$ be an internally derivable system such that admissibility satisfies:

$$x_t \in X \iff S(\omega_t) = 1.$$

If structure S is not equivalent to any state-transition function $F : X \to X$ (Theorem 1), then no internal derivation can extend a trajectory across an N_1 event (Theorem 2).

Formally:

$$S \not\equiv F \implies \text{Non-derivability across } N_1.$$

Therefore, **non-derivability across rupture is a necessary consequence of the non-dynamical nature of structure**.

Proof

Assume that structure S is not equivalent to any state-transition function defined on X.

Then S cannot be represented as a closed mapping $F : X \to X$, and there exist configurations $\omega \in \Omega$ such that:

$$S(\omega) = 0.$$

By definition of admissibility, $S(\omega) = 0$ implies that the corresponding state does not belong to X. Hence, no successor state exists within the domain on which F is defined.

Any trajectory reaching such a configuration necessarily exits the domain of internal derivability. Consequently, no internal extension of the dynamics can cross the boundary defined by an N_1 event.

Thus, derivability across structural rupture is impossible.

Remark

The converse implication is not asserted. While structural non-equivalence implies non-derivability, non-derivability alone does not uniquely determine the form of the structural constraint. This asymmetry is intentional and sufficient for the framework developed in this paper.

Consequence

This meta-theorem shows that:

- non-derivability is **not an independent postulate**,
- not an epistemic limitation,
- and not a contingent modeling failure.

It is the *formal shadow* of structural non-equivalence.

Wherever structure constrains admissibility without being dynamically representable, rupture and non-derivability necessarily follow.

Editorial Note

This meta-theorem allows the paper to be read in either direction:

- **From ontology to epistemology** (structure ⇒ non-derivability), or
- **From epistemic failure to structural necessity** (non-derivability ⇒ structure).

Interpretation

This theorem establishes that:

- rupture is not a failure of prediction,
- not a consequence of insufficient information,
- and not a stochastic deviation.

It is a **formal limit on derivability itself**, imposed by structural constraints.

No amount of refinement, optimization, or modeling within N_0 can bridge an N_1 boundary.

6.2 Formal Comparison with Chaos and Stochastic Breakdown

To prevent misinterpretation, we explicitly distinguish the notion of structural rupture (N_1) from both chaotic dynamics and stochastic breakdown. Although all three limit predictability, they do so in categorically different ways.

6.2.1 Deterministic Chaos

A chaotic system is characterized by deterministic dynamics with sensitive dependence on initial conditions.

Formally, a chaotic system satisfies:

- A well-defined state space X
- A total transition function $F : X \to X$
- Exponential divergence of nearby trajectories

$$\exists \lambda > 0 \text{ such that } \|F^n(x) - F^n(y)\| \sim e^{\lambda n}$$

Despite unpredictability, the system remains **closed**:

$$\forall x \in X, \quad F(x) \in X$$

Key distinction:
Chaos preserves derivability in principle. Structural rupture does not.

6.2.2 Stochastic Dynamics and Noise

In stochastic systems, the transition rule is probabilistic rather than deterministic.

Formally:

$$F : X \to \mathcal{P}(X)$$

where $\mathcal{P}(X)$ is a probability distribution over X.

Uncertainty arises from randomness, incomplete information, or external perturbations. However, admissibility remains intact:

$$\forall x \in X, \quad \mathrm{supp}(F(x)) \subseteq X$$

Key distinction:
Stochasticity modifies trajectories, but does not invalidate the state space.

6.2.3 Structural Rupture (N_1)

In contrast, an N_1 event is defined by **domain invalidation**.

There exists $x_{t^*} \in X$ such that:

$$F(x_{t^*}) \notin X$$

No refinement of initial conditions, no probabilistic extension, and no noise model can restore closure without redefining the state space itself.

6.2.4 Summary of Formal Differences

PROPERTY	CHAOS	STOCHASTIC BREAKDOWN	STRUCTURAL RUPTURE (N_1)
Dynamics defined	Yes	Yes	No (after t^*)
State space preserved	Yes	Yes	No
Closure holds	Yes	Yes	No
Predictability	Limited	Probabilistic	Impossible
Extension by refinement	Possible	Possible	Impossible
Nature of failure	Epistemic	Epistemic	Structural

6.2.5 Consequence

Structural non-derivability is **not** a limit of knowledge, measurement, or computation.
It is a limit imposed by admissibility itself.

Treating N_1 events as chaotic or stochastic failures constitutes a category error: it attempts to repair a domain invalidation using tools that presuppose domain closure.

- "Chaos preserves trajectories; structure limits existence."

Editorial Remark

This distinction ensures that the framework does not collapse into existing theories of chaos, noise, or complexity, but instead identifies a formally distinct source of explanatory breakdown.

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7. Retrodictive Intelligibility

Define intelligibility as a function of time:

$$I(t) = \begin{cases} 1, & t < t^* \\ \text{retrodictive only}, & t \geq t^* \end{cases}$$

After rupture, explanation proceeds backward rather than forward. History is intelligible only in retrospect.

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8. Implications

This framework applies across domains: evolutionary biology, economics, cognitive systems, and history.

Optimization within N_0 does not guarantee survival across N_1. Persistence is contingent, not earned.

The framework does not deny causality or physical law. It denies *global closure*.

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9. Conclusion

Structure is not an answer to ontological questions. It is the condition under which answers remain possible and finite.

By formalizing structure as a constraint operator, we clarify why prediction fails at collapse, why survival lacks moral implication, and why certainty dissolves at the boundary of derivability.

Response to Referee Comment on the Status of the Corollaries

We thank the referee for the careful reading of the manuscript and for raising the concern that the corollaries presented in Section 4.4 may appear to be narrative or interpretative additions rather than formal results.

We would like to clarify that the corollaries are **not independent claims**, nor heuristic interpretations appended to the main results. Rather, they are **logical consequences that follow necessarily** from the formal statements and proofs of Theorem 1 (Categorical Non-Equivalence of Structure and Dynamics) and Theorem 2 (Non-Derivability Across Structural Rupture).

More precisely:

1. **Dependence on explicit definitions**
All corollaries rely exclusively on previously defined objects: the configuration space Ω, the structural constraint operator $S : \Omega \to 0, 1$, internally derivable systems $N_0 = (X, T, F)$, and rupture events N_1. No new primitives, assumptions, or ontological commitments are introduced at the corollary level.
2. **Logical derivability from Theorem 1**
Theorem 1 establishes that structure cannot be represented as, nor reduced to, any

state-transition function $F : X \to X$. From this non-equivalence alone, it follows formally that:

- structural admissibility cannot be inferred from internal dynamics (Corollary 2),
- admissibility precedes dynamics rather than emerging from it (Corollary 1),
- and structural constraints cannot be encoded as predictive rules within a closed system (Corollary 4).

These results are direct consequences of domain–codomain incompatibility and closure properties, not interpretative extrapolations.

3. **Logical derivability from Theorem 2**
Theorem 2 proves that trajectories defined within an N_0 regime cannot be extended across an N_1 event by internal derivation. From this, it follows necessarily that:

- collapse is not a failure of dynamics but an invalidation of its domain (Corollary 3),
- intelligibility becomes retrodictive after rupture (Corollary 5),
- and any post-rupture dynamics must belong to a distinct regime.

These statements formalize properties of derivability and domain invalidation already proven in the theorem.

4. **Role of the corollaries**
The corollaries do not add new results; they **make explicit what would otherwise remain implicit** in the theorems. Their inclusion serves a clarificatory function: to prevent misinterpretations that would treat non-derivability as an epistemic limitation, a modeling artifact, or a stochastic anomaly.

5. **Formal necessity, not narrative extension**
Removing the corollaries would not weaken the formal core of the paper, but it would obscure its logical consequences. Their presence ensures that readers correctly identify the scope and implications of the theorems without importing external assumptions.

For these reasons, we respectfully maintain that the corollaries are **formal consequences of the established results**, not narrative embellishments, and that their inclusion strengthens the logical transparency and interpretability of the framework.

> *"None of the corollaries relies on empirical assumptions"*

> *Corollary 2 follows from Theorem 1, since any predictive scheme presupposes a representability of S within the codomain of F.*

> *Corollary 4 follows from Theorem 1 (structural non-equivalence) combined with Theorem 2 (impossibility of trajectory extension across N_1).*

> *Each corollary targets a distinct logical object: inferability (Cor. 2), domain closure (Cor. 4), dynamical validity (Cor. 3), or temporal asymmetry (Cor. 5). Apparent similarity reflects shared dependence on structural non-equivalence, not redundancy.*

FORMAL RESULT	IMMEDIATE FORMAL CONSEQUENCE	EXPLANATION OF DEPENDENCE
Theorem 1 (Structure is not equivalent to any dynamic $F : X \to X$)	**Corollary 1** (Pre-Dynamical Status of Structure)	If structure cannot be represented as a state-transition function, it must operate prior to and independently of any dynamics defined on X.
Theorem 1	**Corollary 2** (Impossibility of Structural Prediction)	Since S is not derivable from any F, no amount of internal trajectory information can encode or predict structural admissibility.
Theorem 1	**Corollary 4** (Global Non-Closure)	Non-equivalence implies that closure at the level of X does not extend to Ω, hence global derivability fails even under deterministic local dynamics.
Theorem 2 (Non-Derivability Across Structural Rupture)	**Corollary 3** (Non-Dynamical Nature of Collapse)	If trajectories cannot be extended across N_1, then collapse corresponds to domain invalidation, not to a breakdown of the transition function.
Theorem 2	**Corollary 5** (Retrodictive Asymmetry)	The impossibility of forward extension implies that intelligibility after rupture can only proceed backward from admissible states.
Theorem 1 + Theorem 2	**Meta-Theorem** (Structural Origin of Non-Derivability)	Together, they establish that non-derivability is a necessary consequence of structural non-equivalence, not an independent assumption.

Interpretive Note

All corollaries listed above follow from type incompatibility, domain invalidation, and closure properties already established in the theorems. No corollary introduces new axioms, empirical assumptions, or ontological commitments.

Appendix: Summary of Definitions

- P : physical explanatory space
- M : metaphysical explanatory space
- S : structural constraint operator
- N_0 : internally derivable regime
- N_1 : rupture regime

Appendix A — Concrete Application: Evolutionary Collapse as Structural Rupture

This appendix illustrates how the formal framework applies to a concrete domain without introducing new assumptions. The example is evolutionary dynamics, but the structure generalizes to economic, cognitive, and technological systems.

A.1 Evolutionary Dynamics as an N_0 System

Let X denote the space of viable phenotypic configurations of a population within a fixed ecological niche.

Let $F : X \to X$ represent evolutionary dynamics under selection, mutation, and reproduction over discrete generations:

$$x_{t+1} = F(x_t)$$

As long as environmental constraints remain within tolerance bounds, the system satisfies internal derivability:

$$\forall x_t \in X, \quad F(x_t) \in X$$

Optimization, adaptation, and fitness gradients are well-defined within this regime. This corresponds to an N_0 system.

A.2 Structural Constraints as Environmental Admissibility

Let $\Omega = X \times E$, where E denotes environmental configurations.

Define a structural constraint operator:

$$S : \Omega \to \{0,1\}$$

such that:

- $S(x, e) = 1$ if phenotype x is viable under environment e
- $S(x, e) = 0$ otherwise

Structure does not modify evolutionary dynamics. It determines whether dynamics remain admissible.

A.3 Structural Rupture and Mass Extinction

Suppose an environmental shift $e \to e'$ occurs such that:

$$\forall x \in X, \quad S(x, e') = 0$$

Then the evolutionary dynamic becomes invalid:

$$F(x_t) \notin X$$

This constitutes an N_1 event.

Importantly:

- extinction is not caused by maladaptive dynamics,
- not by insufficient variation,
- and not by stochastic fluctuation.

It is caused by **domain invalidation**.

A.4 Formal Consequences

From Theorem 2 (Non-Derivability Across Structural Rupture), it follows that:

- no evolutionary trajectory within X can be extended beyond the rupture,
- no gradual adaptation guarantees survival across N_1,
- optimization within N_0 provides no protection against structural collapse.

Any post-collapse dynamics necessarily occurs in a different regime:

$$N_0' = (X', T', F')$$

with $X' \neq X$.

A.5 Distinction from Gradual Environmental Change

Gradual environmental variation corresponds to changes in F while preserving admissibility:

$$S(x, e_t) = 1 \quad \forall t$$

Structural rupture corresponds to admissibility loss:

$$S(x, e_{t^*}) = 0$$

No refinement of mutation rates, selection pressure, or population size can restore derivability without redefining the state space.

A.6 Interpretation

This application demonstrates that:

- collapse is structurally discontinuous,
- survival is contingent, not earned,
- and evolutionary success does not imply robustness across structural thresholds.

The same formal pattern applies to financial systems, technological infrastructures, and cognitive regimes.

A.7 Scope Note

This appendix does not introduce empirical claims about specific extinction events. It shows that, given the formal framework, extinction-type phenomena are **structurally unavoidable outcomes**, not failures of evolutionary optimization.

Appendix B — Economic Crises as Structural Rupture

This appendix applies the framework to economic systems and financial crises, showing that systemic collapse corresponds to structural invalidation rather than to chaotic dynamics, stochastic shocks, or mispriced risk.

B.1 Economic Dynamics as an N_0 System

Let X denote the space of admissible economic states of a financial system, including prices, leverage ratios, liquidity levels, and balance-sheet configurations.

Let $F : X \to X$ represent the internal economic dynamics generated by trading, credit creation, interest rates, and regulatory constraints:

$$x_{t+1} = F(x_t)$$

As long as institutional rules, legal frameworks, and market conventions remain valid, the system is internally derivable:

$$\forall x_t \in X, \quad F(x_t) \in X$$

Within this regime, optimization, forecasting, and risk modeling are meaningful. This defines an economic N_0 system.

B.2 Structural Constraints as Institutional Admissibility

Let $\Omega = X \times I$, where I denotes institutional configurations (legal regimes, regulatory frameworks, trust conditions, settlement guarantees).

Define a structural constraint operator:

$$S : \Omega \to \{0, 1\}$$

such that:

- $S(x, i) = 1$ if state x is legally, institutionally, and operationally admissible under i
- $S(x, i) = 0$ otherwise

Structure does not alter price dynamics or incentives. It determines whether the system remains admissible as a financial system.

B.3 Crisis as Structural Rupture (N_1)

Suppose an institutional breakdown $i \to i'$ occurs such that:

$$\exists x_t \in X \text{ with } S(x_t, i') = 0$$

Typical examples include:

- loss of counterparty trust,
- invalidation of collateral assumptions,
- breakdown of clearing or settlement mechanisms,
- suspension of convertibility or liquidity guarantees.

Then the internal dynamics become invalid:

$$F(x_t) \notin X$$

This constitutes an N_1 event.

Importantly, the crisis is **not** caused by:

- volatility,
- mispricing,
- stochastic shocks,
- or chaotic amplification.

It is caused by **loss of admissibility**.

B.4 Formal Consequences

From Theorem 2 (Non-Derivability Across Structural Rupture), it follows that:

- no price trajectory can be meaningfully extended across the crisis,
- no internal risk model can predict or absorb the rupture,
- no optimization within N_0 guarantees persistence.

Any post-crisis dynamics necessarily belongs to a different regime:

$$N_0' = (X', T', F')$$

with altered admissibility conditions and institutional structure.

B.5 Distinction from Volatility and Stochastic Shocks

Market volatility corresponds to large variations within X:

$$S(x_t, i) = 1 \quad \forall t$$

Stochastic shocks modify trajectories but preserve closure.

Structural rupture corresponds to:

$$S(x_{t^*}, i) = 0$$

No refinement of forecasting models, stress tests, or probabilistic assumptions can restore derivability without redefining the economic domain itself.

B.6 Retrodictive Nature of Crisis Explanation

After an N_1 event, economic explanation becomes retrodictive:

- balance-sheet vulnerabilities,
- leverage dependencies,
- institutional fragilities

can be reconstructed **only after** collapse.

This follows directly from Corollary 5 (Retrodictive Asymmetry).

B.7 Interpretation

This application shows that:

- financial crises are not extreme market states,
- not failures of rationality,
- and not tails of a distribution.

They are structural events marking the boundary of admissible economic organization.

B.8 Scope Note

This appendix does not claim to explain specific historical crises. It demonstrates that, given the formal framework, crisis-type phenomena necessarily correspond to structural rupture rather than to internal economic dynamics.

Appendix C — Cognitive Framework Collapse as Structural Rupture

This appendix applies the formal framework to cognitive systems, showing that breakdowns of mental frameworks correspond to structural invalidation rather than to information overload, emotional instability, or stochastic noise.

C.1 Cognition as an N_0 System

Let X denote the space of admissible cognitive states of an agent, including beliefs, inferences, expectations, and action policies.

Let $F : X \to X$ represent internal cognitive dynamics such as belief updating, learning, and decision-making:

$$x_{t+1} = F(x_t)$$

As long as the agent's conceptual framework remains valid, cognition is internally derivable:

$$\forall x_t \in X, \quad F(x_t) \in X$$

Within this regime, reasoning, prediction, and adaptation are meaningful. This defines a cognitive N_0 system.

C.2 Structural Constraints as Conceptual Admissibility

Let $\Omega = X \times C$, where C denotes conceptual frameworks (categories, norms of inference, semantic constraints).

Define a structural constraint operator:

$$S : \Omega \to \{0, 1\}$$

such that:

- $S(x, c) = 1$ if cognitive state x is interpretable within framework c
- $S(x, c) = 0$ otherwise

Structure does not modify cognitive transitions. It determines whether cognition remains intelligible as cognition.

C.3 Cognitive Collapse as an N_1 Event

Suppose a conceptual rupture $c \to c'$ occurs such that:

$$\exists x_t \in X \text{ with } S(x_t, c') = 0$$

Examples include:

- loss of meaning coherence,
- breakdown of inferential norms,
- invalidation of core categories or identity schemas.

Then internal cognition becomes undefined:

$$F(x_t) \notin X$$

This constitutes a cognitive N_1 event.

Crucially, the collapse is **not** caused by:

- emotional intensity,
- cognitive bias,
- insufficient information,
- or random disturbance.

It is caused by **conceptual inadmissibility**.

C.4 Formal Consequences

From Theorem 2 (Non-Derivability Across Structural Rupture), it follows that:

- no internal reasoning process can bridge the collapse,
- no gradual belief revision restores coherence,
- no increase in information resolves the rupture.

Any post-collapse cognition necessarily belongs to a different regime:

$$N_0' = (X', T', F')$$

with altered conceptual admissibility.

C.5 Distinction from Stress, Noise, and Cognitive Bias

Cognitive stress and bias correspond to distortions within X:

$$S(x_t, c) = 1 \quad \forall t$$

Stochastic disturbances affect belief trajectories but preserve interpretability.

Structural rupture corresponds to:

$$S(x_{t^*}, c) = 0$$

No refinement of heuristics or emotional regulation can restore derivability without reconstructing the conceptual framework itself.

C.6 Retrodictive Nature of Meaning Reconstruction

After a cognitive N_1 event, intelligibility is retrodictive:

- prior assumptions,
- latent inconsistencies,
- unexamined categories

become identifiable **only after** collapse.

This follows directly from Corollary 5 (Retrodictive Asymmetry).

C.7 Interpretation

This application shows that:

- mental breakdowns are not extreme cognitive states,
- not failures of intelligence,
- and not reducible to pathology.

They mark the boundary of conceptual admissibility.

C.8 Scope Note

This appendix does not address clinical diagnosis or individual psychology. It demonstrates that, within the formal framework, cognitive collapse is a structural phenomenon, not a dynamical malfunction.

Final Summary Table — Structural Pattern Across Domains

DOMAIN	STATE SPACE X	STRUCTURAL CONSTRAINT S	N_1 EVENT (STRUCTURAL RUPTURE)
Physics / Metaphysics	Admissible physical–metaphysical configurations	Admissibility of explanatory configurations	Invalidation of explanatory coherence (no extension of description possible)
Evolutionary Systems	Viable phenotypic configurations	Environmental viability	Extinction via loss of viability under new environment
Economic Systems	Admissible financial states (prices, leverage, liquidity)	Institutional and legal admissibility	Financial crisis via institutional breakdown
Cognitive Systems	Interpretable cognitive states (beliefs, inferences)	Conceptual admissibility	Collapse of meaning / framework invalidation
General N_0 Systems	Internally derivable states	Structural admissibility	Domain invalidation: $F(x_{t^*}) \notin X$

Reading Guide (Formal, Not Interpretive)

- X specifies the **domain on which internal dynamics are defined**.
- S specifies **admissibility conditions external to dynamics**.
- N_1 denotes **structural invalidation**, not dynamical failure.

Across all domains:

$$\text{Optimization within } X \;\not\Rightarrow\; \text{Persistence across } N_1$$

Final Formal Claim

This table shows that the framework does not introduce domain-specific mechanisms.
Each application instantiates the same abstract structure:

- internal derivability (N_0),
- external admissibility (S),
- and non-derivable rupture (N_1).

Differences between domains concern **what occupies X**, not **how collapse occurs**.